

Replication Package for: A Welfare Analysis of Policies Impacting Climate Change

Robert W. Hahn* Nathaniel Hendren[†]
Robert D. Metcalfe[‡] Ben Sprung-Keyser[§]

March 4, 2026

1 Overview

The replication package allows anyone to re-create all figures, tables, and numbers reported in the text using Stata. There is also a series of input assumptions that users can change to generate new MVPF estimates. The replication package is structured as follows. Section 2 will contain the Data Availability Statement. Section 3 will list the necessary computational requirements for running the replication packet. Section 4 will outline the folder structure for the code and data used for replication and will discuss the key code files and data files in greater detail. Section 5 will detail the steps for running the replication package and includes instructions for changing the code to reflect alternative specifications. The code takes significant time (> 2 days) to run the entire replication package, including generating all the estimates with bootstrapped standard errors.

*University of Oxford and Carnegie Mellon University

[†]Massachusetts Institute of Technology and Policy Impacts

[‡]Columbia University and NBER

[§]Wharton School, University of Pennsylvania and Policy Impacts

2 Data Availability Statement

Datasets and citations for data sources used in the analysis are provided in the following locations:

- data/1_assumptions/policy_category_assumptions_MASTER.xlsx
- data/1_assumptions/evs/Record of EV Data Sources.xlsx
- data/1_assumptions/default_assumptions_toggles_vMAIN.xlsx
- all files in data/2a_causal_estimates_papers
- all files in data/1_assumptions/evs
- all files in data/1_assumptions/marginal_damages

2.1 Statement about Rights

- I certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.
- I certify that the authors of the manuscript have documented permission to redistribute/publish the data contained within this replication package.

All data are publicly available, with the following exceptions:

- Cox Automotive/KBB EV sales data (via Statista): requires paid institutional subscription.
- AP3 model data: publicly available upon request; see website.

2.2 Data Sources

Table 1
Data Sources in policy_category_assumptions_MASTER.xlsx

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
Consumer Price Index (CPI-U)	cpi_index	BLS via FRED	Yes	.xlsx	Data are publicly available. Series CPIAUCSL downloaded from FRED, Federal Reserve Bank of St. Louis: https://fred.stlouisfed.org/series/CPIAUCSL . No registration required.
Driving and Fuel Parameters	driving_parameters	Various	Yes	.xlsx	Parameters drawn from multiple published sources. Key inputs from Small and Van Dender (2007), U.S. DOE (2020), and Masnadi et al. (2018). All sources are published journal articles or government reports; see sheet for individual parameter citations.
Ethanol Assumptions	ethanol_assumptions	Various	Yes	.xlsx	Parameters drawn from fueleconomy.gov (publicly available at https://fueleconomy.gov), Hubbard, Anderson, and Wallington (2014), and Lee, Kwon, and Wang (2021). No registration required.
Electric Vehicle Parameters	evs	Various	Yes	.xlsx	Parameters from Greene and Leard (2023, working paper), U.S. Treasury average effective tax rates (publicly available at https://home.treasury.gov), U.S. Energy Information Administration, and author assumptions.
Wind Energy Parameters	Wind	Various	Yes	.xlsx	Parameters drawn from multiple published sources: DOE 2021 Land-Based Wind Market Report, USGS wind turbine capacity data, DSIRE renewable energy database, CRS Report R43453, Kay and Ricks (working paper), and DOE lifecycle emissions data. All sources are publicly available government reports or working papers; see sheet for individual parameter citations and URLs. No registration required.

Continued on next page

Table 1 continued

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
Solar Energy Parameters	Solar	Various	Yes	.xlsx	Parameters from NREL cost benchmarks (publicly available at https://www.nrel.gov/docs/fy22osti/80694.pdf), PVWatts Calculator (https://pvwatts.nrel.gov), and NREL solar market data (https://www.nrel.gov/solar/market-research-analyses/solar-installed-system-cost.html). Parameter from Way et al. No registration required.
Weatherization Parameters	WAP	Various	Yes	.xlsx	Parameters from author assumptions, U.S. Treasury average effective tax rates, and U.S. Energy Information Administration. See sheet for individual sources.
Vehicle Retirement Parameters	car_rebate	Hoekstra, Puller, and West (2017)	Yes	.xlsx	Parameters from Hoekstra, Puller, and West (2017), published in <i>AEJ: Applied Economics</i> . Author assumptions documented in sheet.
Gasoline Prices (Monthly)	gas_prices	EIA	Yes	.xlsx	Data are publicly available from the U.S. Energy Information Administration: https://www.eia.gov/dnav/pet/hist/leafhandler.ashx?n=p&s=emm_epm0_pte_nus_dpg&f=m . No registration required.
Gasoline Prices (Annual, Historical)	gas_data_early	DOE	Yes	.xlsx	Data are publicly available from U.S. Department of Energy: https://www.energy.gov/eere/vehicles/fact-915-march-7-2016-average-historical-annual-gasoline-pump-price-1929-2015 . No registration required.
Gasoline Consumption (Monthly)	gas_consumption	EIA	Yes	.xlsx	Data are publicly available from EIA: https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=P&s=MGFUPUS1&f=M . No registration required.
State Gasoline Tax Rates	gas_tax_rates_state	DOT/FHWA	Yes	.xlsx	Data are publicly available. For 1970–2000: https://www.fhwa.dot.gov/ohim/onh00/table2.htm . For 2000–2021: https://www.fhwa.dot.gov/policyinformation/statistics/2021/mf205.cfm . For 2022: https://www.fhwa.dot.gov/policyinformation/statistics/2022/mf121t.cfm . No registration required.

Continued on next page

Table 1 continued

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
Federal Gasoline Tax Rates	gas_tax_rates_federal	FHWA	Yes	.xlsx	Data are publicly available from FHWA: https://www.fhwa.dot.gov/policyinformation/statistics/2020/pdf/fe101a.pdf . No registration required.
Fuel Ethanol Overview	ethanol_blend_share	EIA	Yes	.xlsx	Data are publicly available from EIA Monthly Energy Review, Table 10.3: https://www.eia.gov/totalenergy/data/monthly/ . No registration required.
Crude Oil Costs and Gasoline Dealer Prices	crude_markup	EIA	Yes	.xlsx	Data are publicly available from EIA. Landed cost: https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=I000000008&f=A . Refiner acquisition cost and dealer tankwagon prices available from same EIA petroleum navigator. No registration required.
Crude Oil Spot Price	crude_spot_price	EIA	Yes	.xlsx	Data are publicly available from EIA: https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=pet&s=rwtc&f=a . No registration required.
Refinery Production Data	refining_production	EIA	Yes	.xlsx	Data are publicly available from EIA petroleum navigator. Net input: https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=MCRRIUS1&f=M . Net output and motor gasoline output available from same source. No registration required.
Petroleum Refining Emissions	refining_emissions	EPA	Yes	.xlsx	Data are publicly available from EPA Inventory of U.S. Greenhouse Gas Emissions and Sinks: https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks . Local pollutant data from EPA National Emissions Inventory (NEI). No registration required.
Historical Fuel Economy (1957–1974)	fuel_economy_1957_1974	EPA	Yes	.xlsx	Data are publicly available from EPA (1973), “A Report on Automotive Fuel Economy”: https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=9101X9Y0.txt . No registration required.

Continued on next page

Table 1 continued

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
EPA Automotive Trends Data (1975–2022)	fuel_economy_1975_2022	EPA	Yes	.xlsx	Data are publicly available from EPA Automotive Trends Report: https://www.epa.gov/automotive-trends/explore-automotive-trends-data . No registration required.
Fuel Economy Forecasts	fuel_economy_forecast	EIA	Yes	.xlsx	Data are publicly available from EIA Annual Energy Outlook 2023, Table 40: https://www.eia.gov/outlooks/aeo/data/browser/#/?id=50-AE02023 . No registration required.
Gasoline Sulfur Content	sulfur_content_gas	EPA	Yes	.xlsx	Data are publicly available from EPA (2017), “Fuel Trends Report: Gasoline 2006–2016”: https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=P100T5J6.txt . No registration required.
Diesel Sulfur Content	sulfur_content_diesel	EPA/EIA	Yes	.xlsx	Data are publicly available from EIA Today in Energy: https://www.eia.gov/todayinenergy/detail.php?id=20092 . No registration required.
Vehicle Miles Traveled by Age (ICE)	vmt_by_age_ICE	FHWA/NHTS	Yes	.xlsx	Data are publicly available from the 2017 National Household Travel Survey: https://nhts.ornl.gov/de/work/173100724283/173100724283.html . No registration required.
Vehicle Miles Traveled by County	vmt_by_county	EPA AVERT	Yes	.xlsx	Data are publicly available from EPA AVERT v4.1: https://www.epa.gov/avert . No registration required.
California Cap-and-Trade Auction Data	CA_C&T_data	CARB	Yes	.xlsx	Data are publicly available from California Air Resources Board: https://ww2.arb.ca.gov/sites/default/files/2020-08/results_summary.pdf and https://ww2.arb.ca.gov/our-work/programs/cap-and-trade-program/cap-and-trade-program-data#allocation . Annual allowances from CARB. No registration required.
RGGI Cap-and-Trade Auction Data	RGGI_C&T_data	RGGI	Yes	.xlsx	Data are publicly available from the Regional Greenhouse Gas Initiative: https://www.rggi.org/auctions/auction-results/prices-volumes . No registration required.

Continued on next page

Table 1 continued

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
EU Emissions Trading System Data	ETS_C&T_data	EEA	Yes	.xlsx	Allowance data are publicly available from the European Environment Agency: https://sdi.eea.europa.eu/webdav/datastore/public/eea_t_eu-emission-trading-scheme_p_2005-2024_v02_r00 . Price data from World Bank Carbon Pricing Dashboard: https://carbonpricingdashboard.worldbank.org . FRED USD/EUR exchange rate (https://fred.stlouisfed.org/series/DEXUSEU) used for currency conversion. No registration required.
Vehicle Emission Factors (GREET/MOVES)	GREET_data_ldv_gas; GREET_data_ldv_diesel; GREET_data_hdv	Argonne National Lab	Yes	.xlsx	Data are publicly available. Emission factors from Cai, Burnham, and Wang (2013), Argonne National Laboratory, Systems Assessment Division. Report available at: https://greet.anl.gov .
EPA Vehicle Emission Rates	diesel_emissions_ldv; diesel_emissions_hdv	EPA	Yes	.xlsx	Data are publicly available from EPA Office of Transportation and Air Quality. Estimated U.S. average vehicle emissions rates per vehicle by vehicle type. https://www.bts.gov/content/estimated-national-average-vehicle-emissions-rates-vehicle-vehicle-type-using-gasoline-and
Diesel Light-Duty Vehicle Production	diesel_ldv_production	EPA	Yes	.xlsx	Data are publicly available from EPA Automotive Trends Report, Table A-1 (“Explore Trends Detailed Data”): https://www.epa.gov/automotive-trends/explore-automotive-trends-data#DetailedData . Restricted to observations where diesel powertrain is reported. No registration required.
Vehicle Inventory and Use Survey (VIUS)	diesel_mdv_dist; diesel_hdv_dist; diesel_mdv_fuel_econ; diesel_hdv_fuel_econ; diesel_fleet_composition	DOT/Census	Yes	.xlsx	Data are publicly available from the 2021 Vehicle Inventory and Use Survey: https://www.census.gov/programs-surveys/vius/data/tables.html . Published by DOT Bureau of Transportation Statistics and U.S. Census Bureau. No registration required.

Continued on next page

Table 1 continued

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
Diesel Prices and Tax Rates	diesel_prices	EIA; FHWA	Yes	.xlsx	Data are publicly available. Prices from EIA: https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=p&s=emd_epd2d_pte_nus_dpg&f=a . Federal diesel tax from FHWA: https://www.fhwa.dot.gov/policyinformation/statistics/2020/pdf/fe101a.pdf . State tax rates from FHWA: https://www.fhwa.dot.gov/policyinformation/statistics/2021/mf205.cfm . No registration required.
E85 Fuel Prices	e85_prices	AFDC	Yes	.xlsx	Data are publicly available from the Alternative Fuels Data Center: https://afdc.energy.gov/fuels/prices.html . No registration required.
EV Sales by Model and kWh Calculations	ev_kwh_calcs	Various	Yes	.xlsx	Sales data through 2019 from Argonne National Laboratory. Sales data for 2020–2022 from Cox Automotive/KBB via Statista. Statista data require an institutional subscription; access is not publicly available. Raw data included in replication package. MSRP, battery capacity, and efficiency data from publicly available sources (fueleconomy.gov, manufacturer specifications).
EV Counterfactual MPG	ev_cf_mpg; ev_cf_mpg_calcs; new_2011_clean_mpg	Various	Yes	.xlsx	Data from sources listed in Record of EV Data Sources.xlsx and Holland et al. (2016). Fuel economy ratings from fueleconomy.gov (publicly available).
Annual Vehicle Miles Traveled	vmt_total_annual	AFDC	Yes	.xlsx	Data are publicly available from the Alternative Fuels Data Center: https://afdc.energy.gov/data/10315 . No registration required.
Energy Efficient Appliance Rebate Parameters	energy_rebate	Houde and Aldy (2017)	Yes	.xlsx	Parameters from Houde and Aldy (2017), published in <i>AEJ: Economic Policy</i> : https://www.aeaweb.org/articles?id=10.1257/pol.20140383 . Additional parameters from U.S. Treasury and EIA.

Continued on next page

Table 1 continued

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
State Electricity Prices	kwh_price_state	BLS	Yes	.xlsx	Data originally accessed from BLS Average Energy Prices query tool (February 2024). Original URL (https://data.bls.gov/pdq/SurveyOutputServlet) is a dynamic query tool; link may not resolve to original data. Raw data included in replication package.
Natural Gas Residential Prices	ng_price	EIA	Yes	.xlsx	Data are publicly available from EIA: https://www.eia.gov/dnav/ng/NG_PRI_SUM_A_EPG0_PRS_DMCF_A.htm . No registration required.
Natural Gas Citygate Prices	ng_citygate	EIA	Yes	.xlsx	Data are publicly available from EIA: https://www.eia.gov/dnav/ng/ng_pri_sum_a_EPG0_PG1_DMcf_a.htm . No registration required.
Natural Gas Emission Factors	ng_pollutants	EPA	Yes	.xlsx	Data are publicly available from EPA GHG Emission Factors Hub: https://www.epa.gov/climateleadership/ghg-emission-factors-hub . No registration required.
Electricity Market Parameters	electricity_markup	EIA; Treasury	Yes	.xlsx	Public utility share from EIA Today in Energy: https://www.eia.gov/todayinenergy/detail.php?id=40913 . Corporate tax rate from U.S. Treasury. Both publicly available.
Electricity Generation Mix by State (2020)	electricity_share_2020	EPA eGRID	Yes	.xlsx	Data are publicly available from EPA eGRID Data Explorer: https://www.epa.gov/egrid/data-explorer . No registration required.
REPEAT Grid Projections	princeton_grid	Princeton REPEAT	Yes	.xlsx	Data are publicly available from Princeton REPEAT Project: https://web.archive.org/web/20240811114328/https://repeatproject.org/results . No registration required.
State-Region Crosswalk	crosswalk_state_region	Author-constructed	Yes	.xlsx	Constructed by authors to crosswalk between Census regions and states for HER nudge estimates. Regional definitions from U.S. Census Bureau. Population data from U.S. Census Bureau population estimates (publicly available at https://www.census.gov/programs-surveys/popest.html).

Continued on next page

Table 1 continued

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
Solar Policy State Population Shares	solar_mix	U.S. Census	Yes	.xlsx	Population data from U.S. Census Bureau. Shares constructed by authors. Publicly available at https://www.census.gov/programs-surveys/popest.html .
Levelized Cost of Electricity Estimates	lcoe_2020_new	EIA; LBNL	Yes	.xlsx	LCOE estimates from EIA Annual Energy Outlook (2015, 2018 editions; publicly available at https://www.eia.gov/outlooks/aeo/) and Lawrence Berkeley National Lab (https://emp.lbl.gov). No registration required.
Electricity Sector Wages by State	wages	BLS	Yes	.xlsx	Data are publicly available from BLS Occupational Employment and Wage Statistics, NAICS 221100: https://www.bls.gov/oes/current/naics4_221100.htm . No registration required.
Transmission and Distribution Losses	transmission_distribution	EIA	Yes	.xlsx	Data are publicly available from EIA Annual Energy Outlook, Table 8: https://www.eia.gov/outlooks/aeo/data/browser/#/?id=8-AE02023 . No registration required.
Home Energy Report Experiments	her_compiled	Allcott (2011)	Yes	.xlsx	Estimates compiled from Allcott (2011), “Social Norms and Energy Conservation,” published in <i>Journal of Public Economics</i> .
Aviation Fuel Prices, Taxes, and Quantities	aviation_prices	EIA	Yes	.xlsx	Data are publicly available from EIA. Aviation gas quantity: https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=MGAUPUS1&f=A . Jet fuel quantity: https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=MKJUPUS1&f=A . Jet fuel spot prices: https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=EER_EPJK_PF4_RGC_DPG&f=A . Aviation fuel taxes: https://www.eia.gov/petroleum/marketing/monthly/xls/aviationtaxes.xlsx . No registration required.
Aviation Local Emissions	aviation_local_emissions	EPA NEI	Yes	.xlsx	Data are publicly available from EPA National Emissions Inventory: https://www.epa.gov/air-emissions-inventories/2020-nei-supporting-data-and-summaries . No registration required.

Continued on next page

Table 1 continued

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
State Jet Fuel Tax Rates	state_jet_fuel_taxes	EIA	Yes	.xlsx	Tax data from EIA: https://www.eia.gov/petroleum/marketing/monthly/xls/aviationtaxes.xlsx . Consumption data from EIA State Energy Data System: https://www.eia.gov/state/seds/sep_fuel/html/fuel_jf.html . Publicly available; no registration required.
Commercial Marine Vessel Emissions	marine_emissions	Various	Yes	.xlsx	Data are publicly available. Emission rates from Eastern Research Group (2022), prepared for EPA: https://gaftp.epa.gov/air/nei/2020/doc/supporting_data/nonpoint/CMV/2020NEI_C1C2_Documentation.pdf . Vessel count from ICCT (2023): https://theicct.org/wp-content/uploads/2023/10/Marine-real-world-NOx-Working-paper-20-A4-v6.pdf . Fuel density conversion from Live Bunkers: https://livebunkers.com/heavy-fuel-oil-hfo . Specific fuel consumption from Sustainable Ships: https://www.sustainable-ships.org/stories/2022/sfc . No registration required.
Residual Fuel Oil Prices	residual_fuel_prices	EIA	Yes	.xlsx	Data are publicly available from EIA: https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=p&s=ema_epr_pwg_nus_dpg&f=a . No registration required.
Annual EV Sales	ev_sales_annual	DOE	Yes	.xlsx	Data through 2021 publicly available from DOE: https://www.energy.gov/eere/vehicles/articles/fotw-1327-january-29-2024-annual-new-light-duty-ev-sales-topped-1-million . No registration required.
Wind LCOE and Capacity	wind_lcoe	DOE; LBNL	Yes	.xlsx	Capacity data from DOE: https://windexchange.energy.gov/maps-data/321 . LCOE data from Lawrence Berkeley National Lab: https://emp.lbl.gov/levelized-cost-wind-energy . Both publicly available; no registration required.

Continued on next page

Table 1 continued

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
Certified Emission Reduction Prices and Wind Capacity	CER_prices	Various	Yes	.xlsx	CER price and CDM capacity data from Cael et al. (working paper). LCOE data from IRENA Wind Energy: https://www.irena.org/Energy-Transition/Technology/Wind-energy . Both publicly available.
Additional Data Sources (from OTHER_DATA_CITES sheet)					
Jacobsen et al. (2023) Replication Data	combined_newcars_copy; combined_aes73_copy	Harvard Dataverse	Yes	.dta	Data are publicly available from Harvard Dataverse: https://doi.org/10.7910/DVN/MISSZQ . No registration required. Located in 1_assumptions/gas_vehicles/fleet_emissions/model_year/input_data
AP3 Model Data	Various files in AP3 Raw Data; md_*_2014_VSL2006; fips_apeep	Tschofen, Azevedo, and Muller (2019)	Yes	Various	Data from the AP3 model. Available at https://nickmuller.tepper.cmu.edu/APModel.aspx and https://github.com/ptschofen/PNAS_SectoralMortality . Access requires registration and approval from the data provider. Raw data included in replication package. Located in 1_assumptions/marginal_damages.
EIA Form 923 Generation Data (2020)	generation_by_plant_2020	EIA	Yes	Various	Data are publicly available from EIA Form 923: https://www.eia.gov/electricity/data/eia923/ . 2020 data used. No registration required. Located in 1_assumptions/marginal_damages.
EIA Form 860 Plant Data (2020)	plant_id_2020	EIA	Yes	Various	Data are publicly available from EIA Form 860: https://www.eia.gov/electricity/data/eia860/ . 2020 data used. No registration required. Located in 1_assumptions/marginal_damages.
Data Sources from Record of EV Data Sources.xlsx					
Ziegler & Trancik Battery Cost Data	battery_costs_zt	Harvard Dataverse	Yes	.xlsx	Data are publicly available from Harvard Dataverse: https://doi.org/10.7910/DVN/9FEJ7C . Ziegler and Trancik (2021). No registration required. Located in 1_assumptions/evs.

Continued on next page

Table 1 continued

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
DOE Battery Pack Costs	battery_costs_doe	DOE	Yes	.xlsx	Data are publicly available from DOE: https://www.energy.gov/eere/vehicles/articles/fotw-1272-january-9-2023-electric-vehicle-battery-pack-costs-2022-are-nearly . No registration required. Located in 1_assumptions/evs.
IEA EV Battery Demand	marg_batt_sales_2016_2022_statista	IEA	Yes	.xlsx	Data are publicly available from the IEA Global EV Data Explorer: https://www.iea.org/data-and-statistics/data-tools/global-ev-data-explorer . No registration required. Located in 1_assumptions/evs.
EV Sales by Model	ev_sales_by_model	Argonne National Lab	Yes	.xlsx	Data are publicly available from Argonne National Laboratory: https://www.anl.gov/es/light-duty-electric-drive-vehicles-monthly-sales-updates . No registration required. Located in 1_assumptions/evs.
NHTS VMT by State (2017)	vmt_by_state_2017_survey	FHWA/NHTS	Yes	.xlsx	Data are publicly available from the 2017 National Household Travel Survey: https://nhts.ornl.gov . Average annual vehicle miles per vehicle by household state, vehicle type, and fuel type. No registration required. Located in 1_assumptions/evs.
NHTS VMT by Vehicle Age (2017)	vmt_by_age_2017_survey	FHWA/NHTS	Yes	.xlsx	Data are publicly available from the 2017 National Household Travel Survey: https://nhts.ornl.gov . Average annual vehicle miles per vehicle by vehicle age, vehicle type, and fuel type. No registration required. Located in 1_assumptions/evs.
KBB Electrified Vehicle Sales (Q4 2021)	Q4-2021-Kelley-Blue-Book-Sales-Report Electrified-Light-Vehicles	Kelley Blue Book	Yes	.pdf	Report published by Kelley Blue Book / Cox Automotive Inc. (2021). PDF reports are publicly available at https://www.coxautoinc.com ; no registration required. Located in 1_assumptions/evs.
KBB EV Sales Reports (Q1–Q4 2023)	kbb_ev_sales_q[1-4]2023	Kelley Blue Book	Yes	.pdf	Reports published by Kelley Blue Book / Cox Automotive Inc. (2023–2024). PDF reports are publicly available at https://www.coxautoinc.com ; no registration required. Specific report URLs in source file. Located in 1_assumptions/evs.

Continued on next page

Table 1 continued

Data Name	Sheet(s)	Source	Provided	Format	Data Availability Statement
HEV Sales by Model (1999–2019)	10301_hev_sale_2-28-20	Argonne National Lab	Yes	.xlsx	Data are publicly available from Argonne National Laboratory via the Alternative Fuels Data Center: https://afdc.energy.gov/data/10301 . No registration required. Located in 1_assumptions/evs.
U.S. Census Population Estimates (2000–2019)	pop_by_state_2010_2019; pop_by_state_2000_2009	U.S. Census Bureau	Yes	.xlsx	Data are publicly available from U.S. Census Bureau, Population Division: https://www.census.gov/programs-surveys/popest.html . Annual and intercensal estimates. No registration required. Located in 1_assumptions/evs.
Cole et al. (2023) Replication Data (LDV)	LDV_out_17ihs_I4.dta	AEA Repository	Yes	.dta	Replication data from Cole et al. (2023). Original replication package available at https://www.openicpsr.org/openicpsr/project/190581/version/V1/view . No registration required. Located in 1_assumptions/evs.
Data Sources from default_assumptions_toggles_vMAIN.xlsx					
Social Cost of Greenhouse Gases (SC-CO ₂ , SC-CH ₄ , SC-N ₂ O)	Sheets: 76, 117, 193, 337, 1367	IWG	Yes	.xlsx	Social cost of greenhouse gas estimates from Interagency Working Group on Social Cost of Greenhouse Gases (2021), “Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide.” Report available at https://bidenwhitehouse.archives.gov/wp-content/uploads/2021/02/TechnicalSupportDocument_SocialCostofCarbonMethaneNitrousOxide.pdf . No registration required.
AP3 Model Marginal Damages (Unweighted & Weighted)	AP3Model_unweighted; AP3Model_weighted	Tschofen, Azevedo, and Muller (2019)	Yes	.xlsx	Marginal damage estimates from the AP3 model. Available at https://nickmuller.tepper.cmu.edu/APModel.aspx . Access requires registration and approval from the data provider. Raw data included in replication package.

3 Computational Requirements

3.1 Software Requirements

The replication package requires the following programs to be installed:

- Stata (code was last run with version 19.5)
- Mathematica (code was last run with version 14.2.1)
 - Note: The code will not run to completion if Mathematica is not properly installed. Only the ‘no learning by doing’ specification can be run without Mathematica. Directions for how to turn learning by doing off are in the masterfile.do.
- Matlab (code was last run with version R2024a)

3.2 Controlled Randomness

Random seed is set on line 74 in wrapper_metafile.do.

3.3 Memory, Runtime, and Storage Requirements

- Approximate time needed to reproduce the analyses on a standard 2025 desktop machine: 80 Hours
- Approximate storage space needed: 220 MB

Details: The code was last run on a 2021 Macbook Pro (M1 Max) with 64GB of RAM.

4 Description of programs/code

This list describes all of the folders besides the data folder.

4.1 Code Structure

- masterfile.do
 - Wrapper that runs through the code files below, generating all figures, tables, and in-text numbers.
- ado
 - Stores all of the ado files, which wrapper/macros.do calls in except the cost_curve_masterfile.ado
- bootstrapping

- Stores the main file that implements bootstrapping.do along with all the files called in bootstrapping.do
- calculations
 - Stores pre-processing files called in macros.do to construct environmental externalities for each policy
- cost_curve
 - Stores the ado file used to calculate cost curve externalities
- data_cleaning
 - Stores pre-processing files used to clean electric vehicle-related data
- figtab
 - Stores all the do files that are needed to create the tables and figures in the paper
- policies
 - The Harmonized subfolder stores all of the policy do-files the main MVPF sample uses
 - The Robustness subfolder stores policy do-files only used for robustness analysis
- publication_bias
 - Stores all the do files that are needed to create the publication bias-corrected version of the estimates
- wrapper
 - Stores the main do files called in masterfile.do, including macros.do, metafile.do, and prepare_causal_estimates.do
 - Stores the do files used to create all the main and appendix figures and tables
 - Stores robustness.do, which creates all the robustness numbers reported in the text

4.2 Explanation of Important Code Files

Macros.do This file, stored in the wrapper subfolder, calculates all of the externalities used for each policy’s MVPF. Therefore, any change in specifications requires the user to re-run macros.do. If you are running through masterfile.do, it will automatically call macros.do. Macros.do creates the electricity, natural gas, and vehicle-related externalities.

Policy do files In the code/policies/harmonized subfolder, there is a separate do file for each policy used in the analysis. For a mapping of the policy do-file name to the policy name reported in Table One, see files/policy_details_v3.xlsx.

You cannot run each policy do-file alone. Instead, the masterfile must call it. To run an individual policy, a user can change the `all_programs` positional argument in the masterfile to the name of the specific policy. Each policy loads in the causal effect estimates from the relevant paper(s). These may be the adjusted causal estimates when run externally for bootstrapping or publication bias. Next, the policy do-files calculate each policy's WTP and net government cost. The do-files call on globals, datasets, and ado files introduced in `macros.do` to do this. Finally, the last section computes the MVPF as the ratio of the WTP and cost and stores all the relevant MVPF components as globals. Many do-files also store components you can use to create policy waterfall graphs if desired.

Prepare Causal Estimates Before estimating MVPFs and bootstrapping them, we prepare bootstrap draws of the causal estimates that go into each MVPF calculation. The file `prepare_causal_estimates.do` does this by looping over all the policies in our sample. For each policy, it finds the corresponding causal estimates file in `causal_estimates/uncorrected`, which details point estimates and standard errors. (In cases where standard errors are unavailable, we use t-statistics or p-values to back them out. Where the paper only reports significance levels, we draw uniformly over the range of possible associated p-values. We record the standard error as 0 in cases where the authors do not report anything. We do not use policies with missing standard errors when calculating standard errors.)

The file details the assumed correlational structure between estimates in the column `"corr_direction."` This variable defines blocks in the correlation matrix between estimates: we indicate blocks by different base numbers, and the sign determines the correlation direction. E.g., if we have four variables with `corr_directions` 1, -1, 2, 2 respectively, then 1 and 2 are perfectly negatively correlated, 3 and 4 are perfectly positively correlated, and 1 and 2 are uncorrelated with 3 and 4. We set these correlations to maximize the width of our confidence intervals where estimates are from the same sample. Given point estimates, standard errors, and a correlation matrix, we draw from the implied joint normal distribution 1000 times, generating the 1000 sets of estimates corresponding to our 1000 bootstrap estimates of the MVPF.

Bootstrapping To get confidence intervals for the MVPF estimates, we bootstrap standard errors. This process occurs in `bootstrapping.do`. Since learning-by-doing takes significantly more run time, we treat policies with and without learning-by-doing separately. For policies without learning-by-doing, we run the metafile with 1000 replications. The causal estimates are created in `prepare_causal_estimates` and fed into `bootstrap_wrapper`. For policies with learning-by-doing, we loop through a series of causal estimates to understand the functional form between the causal estimate and the resulting MVPF. Once we have estimated this functional form, we use it to create the bootstrapped MVPFs without running the metafile on 1000 replications. This entire process occurs in `bootstrapping.do`.

Publication Bias To estimate MVPFs corrected for publication bias, we run `pub_bias_wrapper`, which calls four nested do-files. `prep_and_run_matlab` estimates the level of publication bias in our sample of environmental policies. `cdf_plot` generates the implied CDFs from our method of estimating publication bias and from that of Andrews and Kasy (2019), both compared to the empirical CDF of the (absolute value of the) t-stats in our data. `heuristic_graphs` generates the "funnel plot" of the standard errors for the studies in our sample against the corresponding point estimates. Lastly, `replace_w_corrected` corrects the

point estimates of the papers in our sample and edits Excel sheets that the code uses in MVPF calculations.

Robustness This file, stored in the wrapper subfolder, calculates the numbers reported in the text that we do not directly take from a table or figure. These numbers are outputted to the 4_results subfolder and saved as robustness.xlsx.

Figure & Table Creation We store the wrapper files that create all the tables and figures in the wrapper subfolder as tables.do, figures.do, appendix_tables.do, and appendix_figures.do. Each do-file has nested files in the figtab folder, which construct the individual figures and tables. All the tables and figures are stored in 5_graphs and 6_tables, respectively.

4.3 Data Structure

This list includes all the subfolders of the data folder.

- 0_log
 - After each run of the metafile, a new log file will be created and stored here
- 1_assumptions
 - Stores all assumptions used in the externality calculations.
- 2a_causal_estimates_papers
 - This file stores the causal estimates used in each policy analyzed. The name of the spreadsheet corresponds to the name of the policy do-file.
- 2b_causal_estimates_draws
 - This folder stores the bootstrapped causal estimates that are needed when calculating the bootstrapped standard errors. Each metafile run will produce a new subfolder named using the run's timestamp and the run's name. If the reps argument passed into the metafile is set to 0, the rest of the code will only use the point estimate of the causal estimate for each policy.
- 3_bootstrap_draws
 - Similar to 2b_causal_estimates_draws, each metafile run will produce a new subfolder that is named using the run's timestamp and the run's name. Instead of storing the bootstrapped causal estimates, this folder will store the bootstrapped MVPF components.
- 4_results
 - Similar to the folders above, each metafile run will produce a new subfolder in 4_results that is named using the run's timestamp and the run's name. This folder will have the main compiled results from the run. In other words, this file will store the MVPF along with the components

used to construct the MVPF for each policy included in the run. If the same specification have been run more than once, the figure, table and robustness code are set up to use the most recent timestamp.

- 5_graphs
 - This folder stores the main and appendix figures reported in the paper. It also stores some processed data used to create the figures.
- 6_tables
 - This folder stores the main and appendix tables reported in the paper. It also stores some processed data that the code uses to create the tables.
- 7_publication_bias
 - Stores some data used to create the publication bias results.

4.4 Explanation of Important Data Files

Policy Assumptions Masterfile The assumptions that we use to calculate the environmental and fiscal externalities are primarily pulled from the 1_assumptions/policy_category_assumptions_MASTER.xlsx spreadsheet. We list the sources for the data pulled from the spreadsheet on the first tab.

Program Specific Causal Estimates The causal estimates used for each policy are stored in 2a_causal_estimates_p The names of the spreadsheets correspond to the name of each policy do-file. On the raw_data tab of each spreadsheet, we list where we are pulling the estimate in the paper, the link to the paper PDF, and the date we accessed it. We also show any transformations to the causal estimate before it enters the policy do-file.

5 Instructions to Replicators

To create all of the figures, tables, and numbers reported in the paper, the user only has to run the masterfile. In addition, prior to running alternative specifications or smaller sections of the code, the user must first run the masterfile all the way through. Once you run the masterfile, the results will populate in the subfolders of the data folder as described in Section 4.3. Directions are below:

5.1 Initial run/replicating all results

- Confirm that Mathematica and MATLAB have been downloaded to your computer.
- Open masterfile.do in the code folder.
- Make sure the three globals (rerun_data, bootstraps, run_pub_bias) on lines 14-16 are set to “yes”.

- Set the user-specific file paths on lines 41-47:
 - Change the XX on lines 41 and 42 to the username where the file is stored on your computer. Remove the “C:” if running on a Mac.
 - Make sure that the Dropbox and Github global on lines 43 and 44 point to the data folder and the general Github folder, respectively.
 - If using a Mac, add the file location of Wolfram Alpha on line 46.
- Run masterfile.do from top to bottom.

5.2 Rerunning sections of the code/changing run specifications

- Recreating figures, tables, and robustness numbers:
 - Set the three globals on lines 14-16 to “no” and run the masterfile.
 - Figures, tables, and the robustness numbers are exported to the graphs subfolder, table subfolder, and to an Excel file in 4_results folder, respectively.
- Rerunning the data used for the analysis, bootstraps, or publication bias results.
 - To rerun the data without rerunning bootstraps or publication bias, in the masterfile, keep rerun_data as “yes” and change bootstraps and run_pub_bias to “no”.
 - Adjust accordingly for only rerunning bootstraps or publication bias.
- Changing specifications:
 - Lines 84 - 195 of the masterfile determine which specifications you want for each run. Currently, the masterfile will run all nine specifications used in the paper. Here are the directions for altering these specifications:
 - * Alter the mode to baseline by changing “current” to “baseline” ¹.
 - * Change the second metafile.do argument in each of the seven runs to alter the social cost of carbon (SCC). Currently the masterfile uses an SCC of 193, 76, 337, and 1367.
 - * The third, fourth, and fifth metafile argument decides if learning by doing, include profits, and include savings is turned on or off. These can all be changed from “yes” to “no” and vice versa.
 - * In the sixth argument, “all_programs” can be changed to a specific list of policies.
 - * The seventh argument is the number of reps, and should always be set to zero as bootstrapping occurs in the next section of the masterfile.
 - * The eighth argument is the name of the run, which can be altered as well.

If you are regenerating all the data and the bootstraps, you can expect this code to take over 72 hours to run. Once finished, the resulting dataset will be populated in the 4_results folder with the timestamp and name

¹Current references the 2020 specification, and baseline references the in-context specification. Most of the results reported in the paper use the current specification.

of the run. The code stores the bootstrapped MVPFs and category averages in 5_graphs/figures_data with the v3 file name. Note that the figures, tables, and robustness numbers use the most recent version of the run that matches the current run name. If the name of the run is altered, the corresponding code in the figtab will need to be updated as well.

6 List of tables and programs

All main figures are produced by wrapper/figures.do, appendix figures by wrapper/appendix_figures.do, main tables by wrapper/tables.do, and appendix tables by wrapper/appendix_tables.do.

Table 2
Figure production programs

Figure #	Called Program	Output file
Main Figures		
Figure 1	waterfalls_rep.do	waterfall_muehl_efmp_current.png
Figure 2a	waterfalls_rep.do	waterfall_hitaj_ptc_current.png
Figure 2b	wind_solar_paper.do	Fig_2b_wind_elasticities.png
Figure 3a	waterfalls_rep.do	waterfall_pless_ho_current.png
Figure 3b	wind_solar_paper.do	Fig_3b_solar_elasticities.png
Figure 4	mvpf_plots.do	mvpf_subsidies_Fig4_scc193.png
Figure 5	mvpf_plots_nudges.do	corr_plot_nudges.png
Figure 6	waterfalls_rep.do	waterfall_small_gas_lr_current.png
Figure 7	mvpf_plots.do	mvpf_subsidies_Fig7_scc193.png
Figure 8	mvpf_plots.do	mvpf_subsidies_Fig8_scc193.png
Appendix Figures		
Appendix Figure 1a	lbd_graphs_rep.do	lbd_wind.png
Appendix Figure 1b	lbd_graphs_rep.do	lbd_solar.png
Appendix Figure 1c	lbd_graphs_rep.do	lbd_batt.png
Appendix Figure 2a	connected_externalities_driving.do	vehicle_externalities.png
Appendix Figure 2b	stacked_elec_externalities.do	Ap_Fig_2b_stacked_area.png
Appendix Figure 2c	grid_externality_region.do	Ap_Fig_2_map.png
Appendix Figure 3a	mvpf_plots_add.do	mvpf_comparison_Subsidy Robustness.png
Appendix Figure 3b	changing_grid.do	changing_grid_with_scatter.png
Appendix Figure 4	mvpf_plots.do	mvpf_intl_Ap_Fig4_split_with_CIs.png
Appendix Figure 5	bevs_non_marginal.do	Ap_Fig_5_non_marginal.png
Appendix Figure 6	contour_plot.do	Ap_Fig6_rebound.png
Appendix Figure 7a	heuristic_graphs.do	funnel_plot_threshold_5_5pct_sg.pdf
Appendix Figure 7b	heuristic_graphs.do	RD_thresh.pdf
Appendix Figure 8a	cdf_plot.do	cdf_plot.pdf
Appendix Figure 8b	cdf_plot.do	cdf_plot_AK.pdf
Appendix Figure 9	mvpf_plots.do	mvpf_subsidies_App_Fig_9_scc193.png
Appendix Figure 10a	regulations.do	reg_bars_cafe_dk_gas_panel1.png
Appendix Figure 10b	regulations.do	reg_bars_cafe_dk_gas_panel2.png
Appendix Figure 11a	regulations.do	reg_bars_cafe_as_gas_panel3.png
Appendix Figure 11b	regulations.do	reg_bars_cafe_j_gas_panel3.png
Appendix Figure 11c	regulations.do	reg_bars_rps_wind_panel3.png

Table 3
Table production programs

Table #	Called Program	Output file
Main Tables		
Table 1	excel_MVPF_tables_condensed.do	Table1_scc193_main.xlsx
Table 2	cost_per_ton.do	Table2_CE_Table_Avg_scc193
Appendix Tables		
Table 1 Full Version	excel_MVPF_tables.do	Table1_scc193_app.xlsx
Appendix Table 1		policy_details_v3.xlsx
Appendix Table 2	lbd_table_rep.do	Appendix_Table1.tex
Appendix Table 3	excel_MVPF_tables.do	App_Table3_scc193_in_context.xlsx
Appendix Table 4	ci_table.do	App_Table4_CI_with_cis_new.xlsx
Appendix Table 5	excel_MVPF_tables.do	App_Table5_scc76.xlsx
Appendix Table 6	excel_MVPF_tables.do	App_Table6_scc337_main.xlsx
Appendix Table 7	excel_MVPF_tables.do	App_Table7_scc1367_main.xlsx
Appendix Table 8	excel_MVPF_tables.do	App_Table8_no_lbd.xlsx
Appendix Table 9	excel_MVPF_tables.do	App_Table9_eu_grid.xlsx
Appendix Table 10	excel_MVPF_tables.do	App_Table10_no_profits.xlsx
Appendix Table 11	excel_MVPF_tables.do	App_Table11_with_savings.xlsx
Appendix Table 12	excel_ce_lbd_tables.do	Table12_CE_Table_All_Policies_with_LBD_scc193
Appendix Table 13	excel_ce_lbd_tables.do	Table13_CE_Table_All_Policies_no_LBD_193
Appendix Table 14	excel_ce_lbd_tables.do	Table14_CE_Table_DWL_193
Appendix Table 15	connected_externalities_driving.do	Appendix Table Gas Externalities.xlsx

References

- Isaiah Andrews and Maximilian Kasy, “Identification of and Correction for Publication Bias,” *American Economic Review*, vol. 109, no. 8, pp. 2766–2794, August 2019.
- Cassandra Cole, Michael Droste, Christopher Knittel, Shanjun Li, and James H. Stock, “Policies for Electrifying the Light-Duty Vehicle Fleet in the United States,” *AEA Papers and Proceedings*, vol. 113, pp. 316–322, May 2023.